DSCI 632 – Applied Cloud Computing

Predicting County-Level Housing Prices Using Zillow Data and Apache Spark

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1. Objective

Housing prices are an important indicator that affects investment decisions and market trends. Large -scale datasets such as Zillow's Housing Value Index provide useful insights into housing market trends. Understanding how housing prices change over time can help policymakers, investors, and analysts make better decisions.

The objective of this project is to analyze county-level housing price trends using the Zillow Housing Value Index (ZHVI) dataset and develop a predictive model to forecast future housing prices. The project uses Apache Spark and PySpark to process large datasets and build scalable machine learning models. This study investigates how well recent price trends can predict short-term future housing prices.

2. Dataset

The dataset covers middle-tier housing prices (33rd–67th percentile) for single-family homes and condos, with seasonally adjusted and smoothed values. Each row represents a county with geographic details such as Region ID, Region Name, State, Metro, and SizeRank, and monthly price columns from January 2000 to December 2025.

For this study, the analysis focuses on five states:

- New Jersey
- New York

- Pennsylvania
- California
- Texas

To reduce missing values and focus on modern housing market trends, we have focused on 2010-2025 marketing values. The dataset was transformed from a wide-time series format into a long format where each row represents a county-month observation.

3. Data Preprocessing

Data Preprocessing was performed using Apache Spark and Pyspark to efficiently handle the large dataset. The following steps were applied:

1. State filtering: Only counties from the five selected states were retained

2. **Time filtering**: Monthly housing price columns from 2010 onwards were selected

3. **Wide-to-long transformation**: The dataset originally stored each month as a separate column. Using Spark's *stack()* function, the data was reshaped into a long time-series format with the following fields:

- RegionID
- RegionName
- State
- Metro
- Date
- ZHVI housing price

4. Missing value: Rows containing missing values were removed to have data consistency.

After preprocessing, the dataset contained 78633 county-month observations across the selected states.

4. Feature Engineering

Feature Engineering was performed using Apache Spark window functions to create time-based features for each county. A window was partitioned by *RegionID* and ordered by date to ensure lag values were calculated independently per county.

- **lag_1**: price from 1 month prior
- **lag_3**: price from 3 months prior
- **lag_12**: price from 12 months prior (same month last year)

The target variable (label) was defined as next month's ZHVI price using the *lead*() function. Rows with missing lag or label values were removed, as the first 12 rows of each county do not have a full lag history.

5. Experimental Setup

The dataset was split using a time-based approach to preserve the temporal structure of the data. Data from 2010 to 2022 was used for training (57,009 rows, 78.34%), while data from 2023 to 2025 was used for testing (15,761 rows, 21.66%).

The regression models were evaluated using Apache Spark ML Pipelines:

1. *Linear Regression*
2. *Gradient Boosted Trees (GBT)*
3. *Random Forest*

All models used lag_1, lag_3, lag_12 as input features. These features were combined into a single feature vector using *VectorAssembler*. Model performance was evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

6. Results

The three models were tested on the dataset, and the results are shown below:

Model	**RMSE**	**MAE**
Linear Regression	$3,120.14	$2,082.42
Gradient Boosted Trees	$73,152.80	$29,117.50
Random Forest	$71,093.77	$23,697.80

Linear Regression clearly performed better than the two tree-based models. The per-state breakdown of Linear Regression errors is shown below:

States	RMSE	MAE
Pennsylvania (PA):	$1,720	$1,340 - Lowest Error
New York (NY):	$2,511	$1,655
New Jersey (NJ):	$2,647	$2,004
Texas (TX):	$2,676	$2,032
California (CA):	$5,741	$3,637 - Highest Error

These results align with the average home prices across states. California had the highest average home price ($429,318) resulting in higher prediction errors, while Pennsylvania had the lowest average home price ($166,470).

7. Discussion

The results suggest that recent historical housing prices are strong predictors of short-term price movements. The lag features (lag_1, lag_3, lag_12) capture recent trends and seasonal patterns, which helped the Linear Regression model predict next-month housing prices with low error. Linear Regression performed better because housing prices usually change gradually over time, creating a simple and consistent relationship between past and future prices. In comparison, the tree-based models (Random Forest and Gradient Boosted Trees) rely on decision splits in the data. These models are better for complex nonlinear relationships, but they may struggle to follow smooth time trends like housing prices, which resulted in larger prediction errors.

8. Conclusion

This project developed an Apache Spark pipeline to analyze and predict county-level housing prices using the Zillow ZHVI dataset. The data was transformed into a time-series format, and lag-based features were created to show recent price trends. Three regression models were evaluated using a time-based train/test split, and Linear Regression achieved the best performance. These results show that recent housing prices are strong predictors of short-term price movements. Future work could include additional variables such as interest rates or economic factors to improve prediction accuracy.